

- 5 The variables x and y are related by the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 4e^{-x}.$$

- (a) Find the value of the constant k such that $y = kxe^{-x}$ is a particular integral of the differential equation. [4]

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- (b) Find the solution of the differential equation for which $y = \frac{dy}{dx} = \frac{1}{2}$ when $x = 0$. [6]

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